

ASIDE: UNIX SIGNALS

A huge and fascinating infrastructure known as **signals** is present in all modern UNIX variants. At its simplest, signals provide a way to communicate with a process. Specifically, a signal can be delivered to an application; doing so stops the application from whatever it is doing to run a **signal handler**, i.e., some code in the application to handle that signal. When finished, the process just resumes its previous behavior.

Each signal has a name, such as **HUP** (hang up), **INT** (interrupt), **SEGV** (segmentation violation), etc; see the manual page for details. Interestingly, sometimes it is the kernel itself that does the signaling. For example, when your program encounters a segmentation violation, the OS sends it a **SIGSEGV** (prepending **SIG** to signal names is common); if your program is configured to catch that signal, you can actually run some code in response to this erroneous program behavior (which can be useful for debugging). When a signal is sent to a process not configured to handle that signal, some default behavior is enacted; for **SEGV**, the process is killed.

Here is a simple program that goes into an infinite loop, but has first set up a signal handler to catch **SIGHUP**:

```
#include <stdio.h>
#include <signal.h>

void handle(int arg) {
    printf("stop wakin' me up...\n");
}

int main(int argc, char *argv[]) {
    signal(SIGHUP, handle);
    while (1)
        ; // doin' nothin' except catchin' some sigs
    return 0;
}
```

You can send signals to it with the **kill** command line tool (yes, this is an odd and aggressive name). Doing so will interrupt the main while loop in the program and run the handler code `handle()`:

```
prompt> ./main &
[3] 36705
prompt> kill -HUP 36705
stop wakin' me up...
prompt> kill -HUP 36705
stop wakin' me up...
prompt> kill -HUP 36705
stop wakin' me up...
```

There is a lot more to learn about signals, so much that a single page, much less a single chapter, does not nearly suffice. As always, there is one great source: Stevens and Rago [SR05]. Read more if interested.